

春二周作业

1.

$$Y_1 = \overline{1 \cdot 1 \cdot 1} = 0 \text{ 低电平}$$

$$Y_2 = 1 + 0 = 1 \text{ 高电平}$$

$$Y_3 = \overline{1 \cdot 0} = 1 \text{ 高电平}$$

$$Y_4 = \overline{0 + 0} = 1 \text{ } \overline{0+1} = 0 \text{ 低电平}$$

$$Y_5 = \overline{1} = 0 \text{ 低电平}$$

$$Y_6 = Z \text{ 高阻态}$$

$$Y_7 = \overline{1 \oplus 0} = 0 \text{ 低电平}$$

$$Y_8 = \overline{(1 \cdot 0) + (1 \cdot 1)} = \overline{0 + 1} = 0 \text{ 低电平}$$

2. $Y_1 = \overline{0 + 0} = 1 \text{ 高电平}$ $Y_2 = \overline{1 \cdot 0} = 1 \text{ 高电平}$

$$Y_3 = \overline{1 + 0} = 0 \text{ 低电平}$$

$$Y_4 = \overline{0 \cdot 1} = 0 \text{ 低电平}$$

3. $R_{LLmax} = \frac{V_{DD} - V_{OH}}{nI_{OH} + mI_{IH}} = \frac{5 - 3.4}{2 \times 100\mu + 5 \times 20\mu} = 5.3 \text{ M}\Omega \text{ } 5.0 \text{ M}\Omega$

$$R_{LLmin} = \frac{V_{DD} - V_{OL}}{I_{OLmax} - m'I_{IH}} = \frac{5 - 0.4}{8 \text{ m} - 5 \times 0.4 \text{ m}} = 176 \Omega$$

$$821.43 \Omega$$

85

100